



CENTUM VP ENGINEERING & MAINTENANCE (VP-EM)

OBJECTIVE: This course enables participant to perform generation of CENTUM VP feedback control functions, operator station display functions, sequence control functions, sequence logic functions and graphic panels using CENTUM VP system.

DURATION : 10 Days

PARTICIPANTS: For instrumentation engineers involved in software generation or modification on the CENTUM VP system.

PREREQUISITE: Participants should have basic knowledge on process measurement and control.

COURSE CURRICULUM :

Day-1	Training registration
	Introduction to process control
	DCS evolution
	CENTUM VP system architecture
	FCS Hardware configuration
	PFCS
	LFCS
	KFCS
	FFCS
	FFCS-L
	FFCS-V
	HIS configuration
	Network details
	Address settings

	CENTUM VP system builder configuration
	Project creation
	Project attribution utility
	Introduction to CENTUM VP FCS simulation S/W
	User login
	IOM creation
	Signal configuration
	Concept of control drawing
	Concept of function block diagram

Day-2	Configuration of feedback control functions Creation of open loop Creation of closed loop
	Introduction to standard display windows System defined windows User defined windows
	Operation of instrument faceplates
	Creation of cascade loop
	Operation of instrument faceplates
	Components of instrument faceplate
	Mode status/ Sub status
	Alarm status & alarm settings
	Practice session

Day-3	Tag importance level setting
	Security level setting
	Alarm priority setting
	Tuning window
	System message banner
	Process alarm window
	System alarm window System status display Message monitor window Process report Historical report Navigator window Tool box
	Control group window
	Overview window
	Trend configuration
	Practice session

Day-4	Creation of control loops Signal selectors Auto selectors
	Control signal splitter (SPLIT)
	RATIO
	FOUT
	Auxiliary blocks SW-33
	Temperature and pressure correction block TPCFL
	Velocity limiter VELLIM
	Arithmetic block
	Practice session

Day-5	Calculation block
	Concept of discrete Inputs and outputs
	Concept of software Inputs and outputs
	Common switches
	Global switches
	Messages
	Switch instruments
	Practice session

Day-6	Configuration of sequence control functions
	Concept of interlocks
	Sequence table
	Logic chart
	Configuration of sequence table
	AND logic
	OR logic
	Configuration of switches in sequence table
	Application examples
	Sequence elements
	Timers
	Counters
	Relational expression
	Mode status change through sequence table
	Alarm status usage in sequence table
	Practical session

Day-7	Message processing through sequence table
	Annunciator
	Operator guide
	Printer message
	Application examples
	Concept of logic chart
	Configuration of logic chart
	Motor control block
	MC-2 block execution using LC
	Practice session

Day-8	Report package
	FCS configuration
	HIS configuration
	Project common engineering
	Practice session

Day-9	Real time graphic generation
	Graphic features and settings
	Application example for graphic generation
	Implementation in graphics
	HIS setup
	Practice session

Day-10	CENTUM VP Hardware maintenance
	Downloading
	CENTUM VP software installation
	Project backup and restoration
	CENTUM VP troubleshooting
	Practice session